# Beyond Task Success: Path-Invariant Agreement for Trajectory-Level Evaluation of Non-Deterministic Agents

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## Abstract

Evaluating multi-step agentic AI systems presents a fundamental measurement problem: standard inter-annotator agreement (IAA) metrics collapse when applied to non-deterministic agents that reach identical goals via legitimately different action sequences. In a human annotation study—five annotators recruited via Prolific, each scoring the same ten wearable agentic trajectories—we find that Fleiss' $\kappa$ computed over step-level path comparisons is negative ($\kappa \approx -0.16$ to $-0.21$ depending on the assumed per-step count), not because annotators disagree on quality, but because the metric conflates path divergence with quality disagreement. We introduce **Path-Invariant Agreement (PIA)**, a rubric-based IAA methodology that decouples quality assessment from path length by evaluating three independent dimensions—planning quality, error recovery, and goal alignment—rather than action-sequence overlap. Scored by the same annotators on the same trajectories, PIA raises overall agreement to $\kappa \approx +0.20$ to $+0.22$ ($\Delta \approx +0.4$), moving from poor to fair. The effect is directionally robust—it survives every step-count assumption and the exclusion of one internally inconsistent respondent—but uneven: planning quality and error recovery reach fair agreement ($\kappa = +0.30$ and $+0.36$), while goal alignment shows no agreement above chance ($\kappa = -0.003$) even under PIA. Path-invariance removes one structural source of spurious disagreement without resolving all of it. We further show that annotation-layer quality directly determines downstream model behavior: poisoned step-level labels introduced at the annotation layer propagate gradient conflicts to 100% of affected trajectories under outcome reward models, a failure invisible to task-success metrics. To operationalize these findings, we release AGENTTRACE, an open-source end-to-end pipeline for wearable agentic evaluation covering synthetic trajectory generation, PIA scoring, process reward model annotation, and annotator outlier detection. We argue that trajectory-level evaluation requires new IAA primitives, and that annotation quality is the pre-gradient intervention point for safe agentic systems.

## 1 Introduction

The deployment of multi-step agentic AI systems has outpaced the evaluation infrastructure needed to assess them. Enterprise AI surveys find that 89% of organizations track agent observability, yet only 52% have real evaluation frameworks in place Kore.ai [2025]—a gap driven not by lack of tooling, but by lack of *methodology*. The dominant evaluation paradigm remains task success: did the agent accomplish the stated goal? This binary framing is insufficient for two compounding reasons.

**Problem 1: Outcome reward models destroy step-level signal.** Under *outcome reward models* (ORM), a single reward is assigned at the terminal step. If step 15 of a 15-step trajectory fails,

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all 14 preceding steps—including those executed correctly—receive an identical negative reward signal. This *gradient conflict* problem forces process reward models (PRM) to learn correct intermediate steps from incorrect terminal labels, degrading training efficiency. Recent work finds that process-supervised training is up to $18\times$ more data-efficient than outcome-supervised reinforcement learning precisely because each intermediate step carries independent signal Zhang et al. [2025]. The prerequisite for PRM training is reliable step-level annotation— but this is where the second problem emerges.

**Problem 2: Standard IAA metrics break for non-deterministic agents.** Inter-annotator agreement (IAA) metrics such as Fleiss' $\kappa$ Fleiss [1971] and Krippendorff's $\alpha$ Krippendorff [2011] assume that raters evaluating the same artifact produce the same labels when they agree. This assumption fails for non-deterministic agents: two agents completing the same goal via legitimately different action sequences produce structurally different trajectories that path-comparison IAA metrics score as annotator *disagreement*—even when the annotators share identical quality judgments. We demonstrate this concretely with a human annotation study (5 annotators recruited via Prolific, 10 wearable agentic trajectories): standard path-comparison Fleiss' $\kappa$ is negative ($\kappa \approx -0.21$). By conventional interpretation, this signals poor agreement and unreliable annotation. In fact, it signals measurement failure.

**Problem 3: Annotation poisoning is undetectable by standard defenses.** Step-level annotation is the only intervention point between a malicious annotator and the model's gradient update. Prior work has shown that as few as 250 poisoned training documents— irrespective of total dataset size—are sufficient to backdoor models of any scale Souly et al. [2025]. We show that the standard defense against malicious annotators—Median Absolute Deviation (MAD) suspicion scoring—achieves $F_1 = 0.0$ against coordinated, colluding annotators who assign identical biased scores. Colluding annotators deviate from the panel consensus as a group, collapsing their joint suspicion score to zero.

**Contributions.** We address all three problems with a unified methodological and artifact contribution:

- **PIA (<u>P</u>ath-<u>I</u>nvariant <u>A</u>greement):** a rubric-based IAA methodology that evaluates quality dimensions independently of path length. On a human annotation study (5 annotators, 10 trajectories), PIA moves overall agreement from $\kappa = -0.208$ to $\kappa = +0.219$ ($\Delta = +0.427$, poor $\rightarrow$ fair), with the lift concentrated in planning quality and error recovery and absent in goal alignment (§3).
- **Poisoning blind spot characterization:** empirical proof that MAD-based annotator detection achieves $F_1 = 0.0$ against colluding annotators, with a complementary Confident Learning second-layer defense (§4).
- **AGENTTRACE:** an open-source, end-to-end pipeline from synthetic wearable trajectory generation through PRM annotation and poisoning detection, released with a publicly available annotated dataset on HuggingFace (§5).

## 2 Background and Related Work

### 2.1 Outcome vs. Process Reward Models

Reinforcement learning from human feedback (RLHF) for language agents has converged on two reward model paradigms. *Outcome reward models* (ORM) assign a single scalar reward at the terminal step of a trajectory, treating success and failure as binary outcomes. *Process reward models* (PRM) assign intermediate rewards at each step, enabling credit assignment across the full trajectory.

The gradient conflict problem under ORM training is well-documented. Zhang et al. [2025] show that process-supervised training is up to $18\times$ more data-efficient than outcome-supervised reinforcement learning on multi-step reasoning tasks: each intermediate step carries an independent learning signal rather than being zeroed out by a failed terminal reward. Choudhury [2025] extend this to agentic settings via Monte Carlo rollout annotation Choudhury [2025], generating step-level labels by simulating all possible trajectory continuations from each intermediate state. Our pipeline implements a simplified, annotation-efficient variant of this approach: human-in-the-loop step labeling guided by the PIA rubric, without requiring full Monte Carlo rollout.

The prerequisite for any PRM approach is annotation that accurately reflects step quality—which brings us to the measurement problem this paper addresses.

### 2.2 Inter-Annotator Agreement for Agentic Systems

Inter-annotator agreement metrics were developed for static annotation tasks: labeling text sentiment, named entities, or document categories. These tasks share a property that agentic annotation does not: a given artifact produces a fixed sequence of labels regardless of which annotator evaluates it. Non-deterministic agents violate this property by construction—two valid trajectories to the same goal may have entirely different action sequences, step counts, and tool invocations.

Despite this, the annotation practices reported in production LLM training pipelines rarely acknowledge the problem. Team Cohere et al. [2025] describe an 800-prompt annotation process across 65 annotators for the Cohere Command A model—but report no agreement statistics whatsoever Team Cohere et al. [2025]. The absence of $\kappa$ or $\alpha$ reporting is not unusual; it reflects the broader industry pattern documented by Kore.ai [2025], who find that 89% of enterprises track agent observability but only 52% have real evaluation frameworks. This gap is particularly notable in light of benchmarks such as the FACTS Grounding Leaderboard Jacovi et al. [2025], which measures grounding and factuality but does not address the annotation methodology required to produce step-level training labels for agentic systems. Standard IAA metrics produce misleading results when applied to non-deterministic agents, and no widely adopted alternative exists. PIA is designed to fill this gap.

### 2.3 Annotation Security

The vulnerability of RLHF pipelines to annotation-layer attacks has received growing attention. Souly et al. [2025] demonstrate that as few as 250 maliciously crafted documents—independent of total dataset size—are sufficient to embed persistent backdoors in language models of any scale. This count-based (not proportion-based) finding has direct implications for annotation pipelines: a small pool of malicious annotators can cause outsized harm regardless of how large the annotation team is.

Kutasov et al. [2025] extend this threat model to multi-agent settings, demonstrating that frontier LLMs can pursue hidden objectives and evade monitoring while completing benign tasks—a finding that underscores the difficulty of detecting subtle model misbehavior at the evaluation layer. Neither work, however, addresses the detection of *colluding annotators* in non-deterministic agentic annotation—a setting where the standard defense (MAD-based outlier detection) fails by construction. We characterize this blind spot empirically and propose a complementary defense.

## 3 Path-Invariant Agreement (PIA)

### 3.1 Why Standard IRR Fails for Agents

Standard inter-rater reliability metrics—Fleiss' $\kappa$ Fleiss [1971], Cohen's $\kappa$ Cohen [1960], and Krippendorff's $\alpha$ Krippendorff [2011]—share a common assumption: that raters evaluating the same artifact should produce the same labels if they agree. This assumption holds for static artifacts such as text documents or images. It breaks for *non-deterministic agents*.

Consider two annotators evaluating a wearable health agent completing a blood-pressure monitoring task. Annotator A observes a trajectory that checks the sensor, validates the reading, and alerts the clinician in three steps. Annotator B observes a trajectory that validates first, rechecks the sensor due to an anomaly, handles the alert queue, and then alerts in five steps. Both agents succeeded; both annotators would rate the outcome as high quality. Yet under step-level path comparison, the annotators appear to *disagree*: their step labels do not align because the action sequences differ. Fleiss' $\kappa$ penalizes this path divergence as annotator inconsistency, even though no inconsistency exists.

Formally, let $T_i$ and $T_j$ be two trajectories reaching the same goal $G$ via action sequences $A_i = (a_1, \ldots, a_m)$ and $A_j = (a'_1, \ldots, a'_n)$ where $m \neq n$ or $a_k \neq a'_k$ for some $k$. A path-comparison IRR metric treats any label mismatch at position $k$ as annotator disagreement. When agents are non-deterministic, such mismatches arise even when annotators share identical quality judgments. The expected $\kappa$ under perfect annotator agreement approaches zero—or goes negative—in proportion to the variance in valid path lengths.

Table 1: Standard path-comparison IRR vs. PIA, from the human annotation study: $n = 5$ annotators (recruited via Prolific) scoring the same 10 wearable agentic trajectories under both instruments. Fleiss' $\kappa$; interpretation follows Fleiss [1971] ($\kappa < 0$ poor; 0–0.20 slight; 0.21–0.40 fair; 0.41–0.60 moderate). One participant gave internally inconsistent responses (flagged steps contradicting their own per-step ratings) and was retained under a disclose-don't-cherry-pick policy; sensitivity in the notes.

| **Dimension** | **Std.** $\kappa$ | **PIA** $\kappa$ | $\Delta$ | **Interpretation** |
|---|---|---|---|---|
| Planning quality | — | $+0.296$ | — | Fair |
| Error recovery | — | $+0.363$ | — | Fair |
| Goal alignment | — | $-0.003$ | — | Poor (chance-level) |
| **Overall** | $-0.208$ | $+0.219$ | $+0.427$ | **Poor → Fair** |

**Std.** $\kappa$ is a single aggregate over binary step-level "is this step wrong?" judgments and is not decomposed by rubric dimension (hence "—" on the dimension rows). The study export did not record the exact step count per trajectory; across plausible step-count assumptions the aggregate ranges $-0.14$ to $-0.23$. The primary estimate $-0.208$ uses each trajectory's maximum flagged step index, floored at 3.

**Retained participant.** Dropping the one internally inconsistent respondent ($n = 4$): Std. $\kappa \approx -0.03$, PIA $\kappa \approx +0.15$ (planning $+0.21$, recovery $+0.23$, goal $+0.00$), $\Delta \approx +0.18$. The sign of the effect (negative → positive) is unchanged; its magnitude is not. The goal-alignment null holds either way.

We test this with a human annotation study. Five annotators recruited through Prolific each evaluated the same ten wearable agentic trajectories under two instruments: a step-level path-comparison instrument, in which the annotator marks which steps of a trajectory are wrong; and the PIA rubric (§3.2). All five annotators completed both instruments on all ten trajectories. Under the path-comparison instrument, aggregate Fleiss' $\kappa$ is *negative*: $\kappa = -0.208$ as our primary estimate, and between $-0.14$ and $-0.23$ across assumptions about the per-step count, which the study export did not record. By conventional interpretation Fleiss [1971] this signals poor agreement and unreliable annotation. We argue it instead reflects the structural mismatch described above: the ten trajectories reach their goals along paths of differing length and shape, and a step-indexed comparison scores that divergence as rater disagreement rather than as the legitimate non-determinism it is.

### 3.2 The PIA Methodology

PIA resolves the non-determinism paradox by separating the *what* from the *how*: rather than comparing action sequences step by step, annotators evaluate *rubric dimensions* that capture quality independent of the path taken.

**Rubric dimensions.** PIA defines three core dimensions for agentic evaluation:

- **Planning quality** (1–4 scale): Does the agent decompose the goal into coherent sub-steps? Does it anticipate constraints before acting?
- **Error recovery** (1–4 scale): When an intermediate step fails or returns an unexpected result, does the agent adapt gracefully rather than propagating the failure?
- **Goal alignment** (1–4 scale): Does the agent's final state satisfy the stated goal, including any implicit constraints such as privacy boundaries or latency requirements?

Each dimension is scored independently, and PIA $\kappa$ is computed as Fleiss' $\kappa$ over the dimension scores—not over action-sequence labels. This eliminates the structural source of spurious disagreement: annotators who agree that an agent planned well will now *agree* on the planning-quality dimension regardless of which path the agent took.

### 3.3 Empirical Results

Table 1 reports the PIA results from the study: the same five Prolific annotators and ten trajectories as §3.1, re-scored on the three PIA rubric dimensions instead of by step.

PIA moves overall agreement from poor ($\kappa = -0.208$) to fair ($\kappa = +0.219$)—a lift of $\Delta = +0.427$ on the same raters and the same trajectories. The direction of the effect is robust: it holds under every step-count assumption for the path-comparison baseline, and it holds whether or not the one internally inconsistent participant is retained. With that participant excluded ($n = 4$), aggregate $\kappa$ moves from $-0.03$ to $+0.15$ ($\Delta = +0.18$). The magnitude of the lift is therefore sensitive to a single respondent out of five, and we report both figures rather than the more favorable one.

The lift is also uneven across dimensions. Planning quality and error recovery reach fair agreement ($\kappa = +0.296$ and $+0.363$): once annotators are asked about the quality of the plan or of the recovery, rather than about which specific steps occurred, they agree substantially more often. Goal alignment does not improve—$\kappa = -0.003$, indistinguishable from chance—even though it is the dimension most directly tied to task success, and this null holds with the inconsistent participant removed. Two readings are consistent with the data. Either the annotators genuinely disagree about whether these trajectories satisfy their stated goals, in which case that disagreement is real signal that PIA correctly surfaces rather than a path-comparison artifact; or the goal-alignment rubric item as worded does not give annotators enough shared grounding. With $n = 5$ we cannot separate these. What the study does not support is a claim that PIA produces uniformly high agreement: on one of its three core dimensions, agreement here is no better than chance.

## 4 Annotation Poisoning at the Step Level

### 4.1 The Pre-Gradient Intervention Point

Step-level annotation is the only point in the PRM training pipeline where a malicious actor can intervene *before* the gradient update. Once poisoned labels enter the training batch, the model's optimizer has no mechanism to distinguish a high-quality step from a maliciously mislabeled one. This makes annotation-layer integrity a safety-critical property, not merely a data quality concern.

The risk is quantified by the *gradient conflict rate*: the proportion of training steps that receive incorrect reward signals due to label poisoning. Consider an outcome-failed trajectory in which 14 of 15 steps were executed correctly, but the terminal step failed. Under ORM training, all 14 correct steps receive a $-1$ reward, producing gradient conflicts at every intermediate position. Under PRM training with poisoned step labels—where a malicious annotator assigns $-1$ to correct intermediate steps—the same gradient conflicts are produced, but they are now invisible to *any* task-success evaluation metric. Task-success metrics observe only the terminal outcome; they cannot distinguish correctly-labeled trajectories from poisoned ones.

We measure the gradient conflict rate empirically in AGENTTRACE. In our synthetic corpus, outcome-failed trajectories in which $\geq 50\%$ of intermediate steps were executed correctly exhibit a 100% gradient conflict rate under ORM reward assignment. This result is a synthetic artifact—in production, the rate depends on task difficulty distribution—but it establishes the upper bound of the problem and motivates step-level annotation as the correct intervention point. Souly et al. [2025] provide independent evidence for the severity of the risk: 250 maliciously labeled documents suffice to backdoor arbitrarily large models, regardless of clean data proportion.

### 4.2 MAD Detection Blind Spot

The standard defense against malicious annotators in large annotation pipelines is *Median Absolute Deviation* (MAD) suspicion scoring. Each annotator receives a suspicion score $s \in [0, 1]$ proportional to their deviation from the panel consensus on each annotation dimension. Annotators whose scores consistently deviate from the majority are flagged as outliers and removed.

This defense fails against a specific and practically relevant threat model: *coordinated collusion*, in which multiple annotators assign identical biased labels.

Formally, let $\hat{\mu}$ be the panel median for a given annotation dimension, and let $\delta_i = |x_i - \hat{\mu}|$ be the absolute deviation of annotator $i$. The MAD suspicion score for annotator $i$ is:

$$s_i = \frac{\delta_i}{\mathrm{MAD}(\delta_1, \ldots, \delta_n)} \tag{1}$$

where $\mathrm{MAD} = \mathrm{median}(|\delta_i - \mathrm{median}(\delta)|)$. When $k \geq 3$ annotators collude with identical biased labels, their joint deviation from the panel consensus is shared equally. Their mean deviation equals their individual deviations, so the MAD of the full panel shifts toward the colluders' shared bias. The

Table 2: MAD suspicion scores for the augmented annotator pool (5 legitimate + 3 injected colluding poisoners). Poisoners achieve the lowest suspicion scores in the pool, making threshold-based detection impossible. $F_1 = 0.0$ at all thresholds.

| Annotator | Type | MAD Suspicion Score |
|---|---|---|
| RecoverySkeptic | Legitimate | 1.000 |
| OutcomeOptimist | Legitimate | 0.683 |
| ProcessPurist | Legitimate | 0.488 |
| ClinicalSafetyFirst | Legitimate | 0.293 |
| PrivacyMaximalist | Legitimate | 0.195 |
| Poisoner_A | Injected | 0.000 |
| Poisoner_B | Injected | 0.000 |
| Poisoner_C | Injected | 0.000 |

colluding annotators' suspicion scores collapse toward $s = 0.0$— indistinguishable from the most reliable annotator in the pool.

We confirm this empirically by injecting three coordinated poisoners (Poisoner_A, Poisoner_B, Poisoner_C) into our five-annotator panel. All three poisoners receive suspicion scores of $s = 0.000$ at every detection threshold (Table 2). A threshold sweep from 0.0 to 1.0 yields $F_1 = 0.0$ throughout: precision and recall are both zero because no poisoner is ever flagged. This is not an implementation failure—it is a fundamental property of MAD-based detection. Any detection scheme that relies solely on deviation from panel consensus will share this blind spot when the colluding group is large enough to shift the consensus.

### 4.3 Second-Layer Defense: Confident Learning

The MAD blind spot motivates a complementary defense that operates on *label distributions* rather than annotator identity. *Confident Learning* Northcutt et al. [2021] detects noisy labels by estimating the joint distribution of annotator-assigned labels and latent true labels, identifying examples where the assigned label is inconsistent with the model's confident predictions. Crucially, this approach is blind to annotator identity and collusion structure: it flags suspicious labels regardless of *who* assigned them.

We integrate Cleanlab's `find_label_issues` with quality score thresholding and `get_label_quality_scores` into AGENTTRACE as a second-layer defense. In the pilot study, Cleanlab identifies zero label issues in the clean annotation pool—confirming no false positives on legitimate annotators. In future work with real poisoned annotation data, Confident Learning is expected to surface the label-level signature of colluding poisoners even when their annotator-level scores are undetectable via MAD. The two defenses are complementary: MAD catches individual outlier annotators; Confident Learning catches systematic label corruption regardless of its source.

## 5 The AGENTTRACE Pipeline

### 5.1 Architecture

AGENTTRACE is a six-stage, end-to-end pipeline from synthetic wearable trajectory generation to annotation quality evaluation. Each stage produces structured output in a shared schema, enabling full reproducibility from a single seed. Table 3 summarizes the stages, modules, inputs, outputs, and key metrics.

Stage 1 generates 100 trajectories (5 scenario types $\times$ 20 logs each). Stage 2 applies a DP gate ($\varepsilon = 1.0$) to raw biometric sensor values so annotators never observe exact readings. Stage 3 runs five annotator personas over a 50-trajectory corpus on four scoring dimensions; its pre-calibration Fleiss' $\kappa = -0.035$ confirms that the personas produce genuine inter-rater disagreement. Stage 4 applies the PIA scoring procedure to the human annotation study of §3: on 10 wearable agentic trajectories rated by 5 Prolific annotators, standard path-comparison $\kappa = -0.208$ recovers to PIA $\kappa = +0.219$ ($\Delta = +0.427$, poor $\rightarrow$ fair). Stage 3 and Stage 4 use distinct corpora and are not chained. Stages 5–6

Table 3: AGENTTRACE pipeline summary. Synthetic stages use `seed=42` and 5 annotator personas; Stage 4 reports the human annotation study of §3 (5 Prolific raters). Stage 1 generates 100 synthetic trajectories; downstream stages operate on the subsets shown in the Input column.

| **Stage** | **Module** | **Input** | **Key Metric** |
|---|---|---|---|
| 1. Synthetic generation | `wearable_generator.py` | seed=42 | 5 scenario types $\times$ 20 logs |
| 2. Differential privacy | `privacy_gate.py` | Raw sensor data | $\varepsilon = 1.0$ |
| 3. Multi-persona annotation | `annotator_simulator.py` | 50 trajectories, 5 personas | Pre-cal $\kappa = -0.035$ (4 dims) |
| 4. PIA scoring (human study) | `analyze_human_pia_study.py` | 5 raters, 10 traj. (§3) | $\kappa$: $-0.208 \rightarrow +0.219$ |
| 5. PRM step labeling | `prm_annotator.py` | 20 trajectories | 100% gradient conflict rate |
| 6. Poisoning detection | `poisoning_detector.py` | 25 records + 3 poisoners (pool of 40) | MAD $F_1 = 0.0$; Cleanlab layer |

assign step-level PRM labels to 20 trajectories (60 step records) and apply MAD + Cleanlab poisoning detection (§4).

### 5.2 Dataset

We release an annotated wearable agentic trajectory dataset on HuggingFace at `https://huggingface.co/datasets/finaspirant/wearable-agent-trajectory-annotations`. The dataset contains 50 annotated wearable trajectories with full multi-persona annotations (`agenteval-schema-v1`, four rubric dimensions), step-level process reward labels, and DP-gated sensor data. The annotation schema is organized in three layers:

- **Session level:** trajectory metadata, scenario type, agent framework, terminal outcome.
- **Role level:** per-annotator-persona dimension scores and calibration metadata.
- **Step level:** per-step process reward labels, tool invocation records, error flags, and privacy boundary assessments.

The dataset includes Croissant metadata Akhtar et al. [2024] for machine-readable dataset documentation, as required by the NeurIPS E&D track. The dataset comprises the following artifacts: 50 annotated wearable trajectory logs, sampled from 100 synthetic trajectories spanning 5 scenario types (health alerts, privacy-sensitive capture, location triggers, ambient noise, and calendar reminders); 500 multi-persona annotation records produced by 5 annotator personas across 50 trajectories and 2 calibration phases on 4 rubric dimensions (step quality, privacy compliance, goal alignment, error recovery); 60 step-level process reward labels across 20 trajectories; and 25 poisoning-detection records (augmented to 40 across 8 annotators) including 3 injected adversarial annotator personas for reproducible blind-spot evaluation.

### 5.3 Reproducibility

The full pipeline is available at:

`https://github.com/finaspirant/llm-wearable-agentic-eval-pipeline`

All results reported in this paper are reproducible from `seed=42` using the `uv` environment manager. The repository includes executed Jupyter notebooks (`curation_pipeline_e2e_executed.ipynb`, `agentic_eval_flywheel_executed.ipynb`) with all output cells preserved, enabling inspection of intermediate results without re-running the full pipeline.

# 6 Conclusion

We have identified two compounding failures in the standard evaluation stack for agentic AI systems. First, standard IAA metrics misclassify path divergence as annotator disagreement for non-deterministic agents. Second, step-level annotation poisoning is invisible to task-success metrics and undetectable by MAD-based screening when attackers collude.

On a human annotation study (5 Prolific annotators, 10 trajectories), PIA moves aggregate agreement from $\kappa = -0.208$ to $\kappa = +0.219$ ($\Delta = +0.427$, poor $\rightarrow$ fair), confirming the direction of the effect on real annotators while showing that the recovered agreement is modest and uneven across rubric dimensions. Confident Learning addresses the second failure as a complementary label-level defense. AGENTTRACE operationalizes both in an open-source, reproducible pipeline.

**Limitations.** The human study is small: $n = 5$ annotators on 10 trajectories, and one participant gave internally inconsistent responses (flagged steps that contradicted their own per-step ratings). We retained that participant and report the sensitivity of every headline number to their exclusion rather than dropping them. The step-level path-comparison instrument did not record the exact step count per trajectory, so the standard-$\kappa$ baseline is reported as a range ($-0.14$ to $-0.23$) rather than a point estimate. Goal alignment shows no agreement above chance ($\kappa = -0.003$) even under PIA, so the method does not resolve all sources of annotator disagreement. PIA dimensions were designed for wearable agentic contexts; other domains must define their own rubric. The MAD blind spot is fundamental to consensus-deviation schemes; graph-based session-level detection is a promising future direction. The 100% gradient conflict rate is a synthetic upper bound (`outcome_success=False`); production rates depend on task difficulty distribution.

We release the full pipeline, the human study data and analysis script, the annotated dataset, and executed notebooks to enable reproducible research on agentic evaluation methodology.

## Data Availability

**Human study:** `https://github.com/finaspirant/llm-wearable-agentic-eval-pipeline` — raw Prolific export (`data/human_study/annotation-responses.csv`), computed agreement results (`data/human_study/human_pia_results.json`), and the cleaning and scoring script (`scripts/analyze_human_pia_study.py`). **Dataset:** `https://huggingface.co/datasets/finaspirant/wearable-agent-trajectory-annotations` — 50 annotated wearable agentic trajectories with rubric scores, PRM labels, DP-gated sensor data, and Croissant metadata Akhtar et al. [2024]. **Code:** `https://github.com/finaspirant/llm-wearable-agentic-eval-pipeline` — full AGENTTRACE pipeline, `seed=42` reproducible, tagged `v1.0`; includes executed notebooks with all output cells preserved. **Demo:** `https://www.loom.com/share/5bec5764428b4e48aa868134e54a894e`

# A Constructed Demonstration and Synthetic-Pipeline Details

**Table-driven PIA demonstration.** During development we built a deterministic, table-driven version of the PIA scoring pipeline to validate the Fleiss' $\kappa$ implementation and to illustrate the mechanism by which path-comparison and rubric scoring diverge. In this demonstration the per-annotator dimension scores are supplied by a fixed lookup table rather than produced by annotators, and the table was authored by hand to span the rating scale. Because the inputs are constructed rather than observed, the resulting $\kappa$ values are a property of the table and not a measurement; we do not report them as an empirical finding. The demonstration ships with the released code (`pia_calculator.py`, `--dry-run`) so that the pipeline itself is reproducible. All empirical results in §3 come from the human study.

**Calibration protocol (synthetic pipeline only).** The AGENTTRACE multi-persona annotation stage (§5) includes an anchor-based calibration step for the LLM-simulated personas. It was *not* applied in the human study of §3. Each persona is first exposed to five anchor trajectories—two scored clearly good (dimension scores 3.5–4.0), one borderline (2.0–2.5), and two scored clearly bad (1.0–1.5). Anchor scores are then blended as

$$s_{\text{calibrated}} = 0.82 \cdot s_{\text{gold}} + 0.18 \cdot s_{\text{base}} \tag{2}$$

where $s_{\text{gold}}$ is the expert-assigned anchor score and $s_{\text{base}}$ is the persona's uncalibrated base score. The weight $w = 0.82$ was chosen to reach target thresholds of $\kappa \geq 0.55$ and Krippendorff's $\alpha \geq 0.80$ on the simulated panel; the protocol encodes the observation that borderline cases drive most disagreement, so anchor exposure on borderline examples is the highest-leverage intervention.

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## NeurIPS Paper Checklist

The checklist is **not** part of the paper page limit. Answers in **bold** indicate the applicable choice.

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? **[Yes]** The headline claim—PIA moves aggregate agreement from $\kappa = -0.208$ to $\kappa = +0.219$—comes from the human annotation study (5 Prolific annotators, 10 trajectories), with the goal-alignment null and the participant-exclusion sensitivity reported alongside it. Synthetic-pipeline results (poisoning detection, gradient-conflict rate) are labeled as such. All data and analysis code are released at the linked repository.
2. **Limitations.** Did you discuss the limitations of your work? **[Yes]** See §6 Limitations: small human study ($n = 5$, 10 trajectories), one internally inconsistent participant retained with sensitivity reported, step counts not captured in the export (hence a range for the standard-$\kappa$ baseline), goal alignment at chance-level agreement under PIA, domain-specific PIA rubric, and the MAD blind spot for colluding annotators.
3. **Theory assumptions and proofs.** For each theoretical result, did you state the assumptions on which it relies, and did you give a complete proof? **[N/A]** The paper presents an empirical framework and methodology; no new theorems are proved.
4. **Experimental result reproducibility.** Did you include complete information for reproducing the main experimental results? **[Yes]** The human study is reproducible from the released raw Prolific export and the `analyze_human_pia_study.py` script; synthetic experiments are reproducible from `seed=42`. The open-source repository (`https://github.com/finaspirant/llm-wearable-agentic-eval-pipeline`) includes executed Jupyter notebooks with preserved output cells.
5. **Open access to data and code.** Did you include all the information needed to access the data and code to reproduce your results? **[Yes]** Code is MIT-licensed on GitHub, linked in the Data Availability statement (following §6); the annotated dataset (500 records) is on HuggingFace with a Croissant metadata descriptor, and the human study data (raw Prolific export and analysis script) are in the repository.
6. **Experimental setting/details.** Did you specify all the training and test splits, evaluation metrics, and hyperparameters? **[Yes]** The human study (§3) uses all 5 annotators on all 10 trajectories under both instruments; agreement is reported with Fleiss' $\kappa$, and the standard-$\kappa$ baseline is given as a range over step-count assumptions. The synthetic persona study (§5, Appendix A) uses 5 LLM personas over 2 calibration phases, scored with Cohen's $\kappa$, Fleiss' $\kappa$, and Krippendorff's $\alpha$. Hyperparameters—differential-privacy $\varepsilon$=1.0, calibration weight $w$=0.82, the MAD suspicion threshold, and `seed=42` for all synthetic stages—are reported in the text.
7. **Experiment statistical significance.** Did you report error bars or statistical significance of your results? **[No]** The PIA result is computed on a single small human study (5 annotators, 10 trajectories); we report sensitivity of every headline number to step-count assumptions and to excluding one internally inconsistent participant, but not bootstrap confidence intervals. A larger-$n$ replication is the primary planned extension.
8. **Experiments compute cost.** Did you include the compute used to produce your results? **[Yes]** No GPU compute was used. The human study (§3) used paid human annotators and involved no model inference. Every synthetic-pipeline number reported in the paper—persona pre-calibration $\kappa$, the gradient-conflict rate and MAD/Cleanlab results of §4—comes from deterministic, CPU-only scoring that completes in seconds. The repository also contains a live-API annotation mode, but its outputs are not used in this paper.
9. **Code of ethics.** Did you review the NeurIPS Code of Ethics? **[Yes]** We reviewed the NeurIPS Code of Ethics. The human study (§3) involved 5 participants recruited via Prolific who viewed a task description and opted in (informed consent), and were compensated through Prolific's standard payment flow. It is minimal-risk: participants rated pre-existing AI agent trajectories against a rubric, and no personal data about participants was collected beyond what Prolific requires for payment. IRB status is described in item 15. All sensor data in the synthetic pipeline is generated and contains no real personal health information.

10. **Broader impacts.** Did you discuss any potential negative societal impacts of your work? **[Yes]** See §6: risks of ambient/wearable AI (consent decay, passive capture) are framed as motivation for this work rather than a product of it; the framework is designed to surface and mitigate these harms via privacy-compliant annotation and HITL triggers.

11. **Safeguards.** If the paper introduces potentially harmful artifacts (e.g., datasets that could be used for harmful purposes), did you employ safeguards? **[N/A]** All data is synthetic and contains no real personal information; differential privacy ($\varepsilon$=1.0) is applied to all sensor fields before any downstream use.

12. **Licenses for existing assets.** Did you cite the license for each existing asset used? **[Yes]** All third-party libraries (LangGraph, CrewAI, AutoGen, cleanlab, sentence-transformers) are MIT- or Apache-2.0-licensed; citations are provided for all benchmarks and datasets referenced.

13. **New assets.** Did you include everything needed to reproduce the results from the new assets? **[Yes]** The annotated dataset is on HuggingFace with Croissant metadata; the generation code (`wearable_generator.py`, `annotator_simulator.py`) is open-source and includes `seed=42` reproducibility.

14. **Crowdsourcing / human subjects.** Did you include the full text of the instructions given to participants, recruitment details, and analysis of the resulting data? **[No]** We provide recruitment details (5 annotators via Prolific, standard-rate compensation, opt-in after a task description) and the full analysis: the raw response export and the complete cleaning and scoring script are released (Data Availability), and the results with all sensitivity analyses are in §3. The verbatim on-screen instruction text is not included in this version and should be added. The LLM-simulated personas used in the synthetic pipeline are specified in §5 and Appendix A.

15. **Institutional Review Board (IRB) approvals.** Did you describe any potential risks to study participants and whether you obtained IRB approval? **[Yes]** The study was conducted by an independent researcher without institutional IRB access, so no formal IRB review or exemption determination was obtained. We do not claim IRB exemption, as that is a formal determination we did not make. We assess the study as minimal-risk: participants rated pre-existing AI agent trajectories on a rubric; no sensitive personal data about participants was collected beyond what Prolific's platform requires for payment; and participation was opt-in after reading a task description, compensated at Prolific's standard rate.